CHAPTER – XI

FUTURE SCOPE

11.1 - Future Scope

The project "Plugging into the Future: An Exploration of Electricity Consumption Patterns" can be extended in several directions to provide deeper insights, better forecasting, and smarter energy management.

1. Real-Time Monitoring

Integrate smart meter and IoT sensor data to monitor electricity usage in real time. This would enable instant alerts, anomaly detection, and dynamic dashboards for consumers and utilities.

2. Advanced Demand Forecasting

Apply machine learning and deep learning models (such as LSTM, GRU, and transformer-based forecasting) to predict short-term and long-term electricity demand with higher accuracy.

3. Personalized Energy Recommendations

Develop recommendation systems that suggest energy-saving actions based on household or organizational consumption patterns, peak usage hours, and appliance behaviour.

4. Integration with Renewable Energy

Extend the analysis to include solar, wind, and battery storage data. This would help evaluate how renewable generation affects overall consumption and grid stability.

5. Predictive Maintenance and Fault Detection

Use consumption anomalies to detect faulty appliances, power theft, or equipment failures before they become critical.